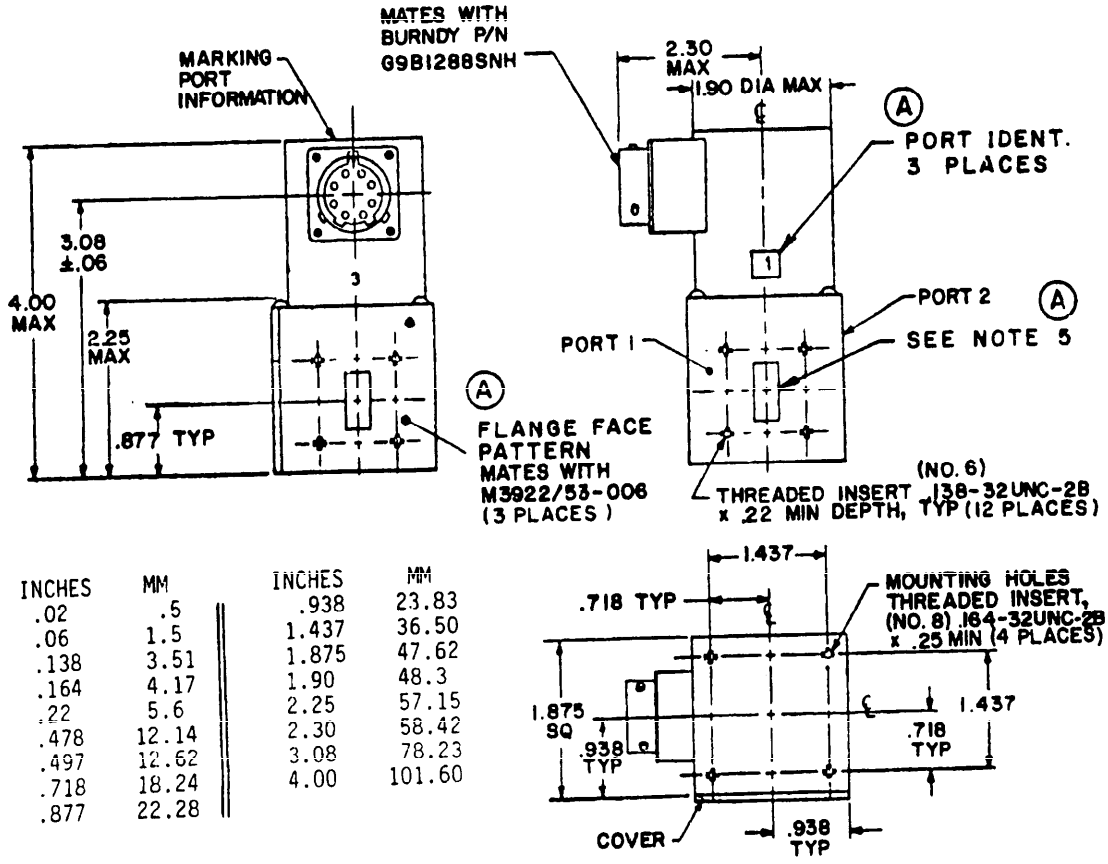
MILITARY SPECIFICATION SHEET

SWITCHES, WAVEGUIDE, 1P2T, FAIL-SAFE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-S-55041.



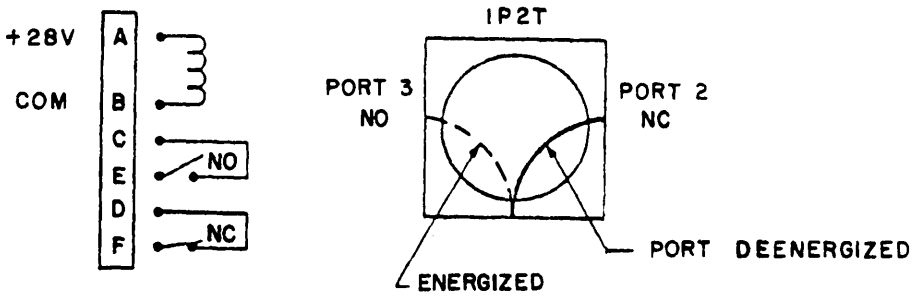
INCHES	MM	INCHES	MM
.02	.5	.938	23.83
.06	1.5	1.437	36.50
.138	3.51	1.875	47.62
.164	4.17	1.90	48.3
.22	5.6	2.25	57.15
.478	12.14	2.30	58.42
.497	12.62	3.08	78.23
.718	18.24	4.00	101.60
.877	22.28		

FIGURE 1. Switch configuration and schematic, part no. M55041/6-001.

(A) denotes changes

FSC 5985

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Auxiliary switches: The following rules shall apply.
Pins D & F connected only when port 1 is connected to port 2
Pins C & E connected only when port 1 is connected to port 3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Round corners of case may be squared.
4. Unless otherwise specified, tolerances are ±.005 (.13mm) for three place decimals and ±.03 (.8mm) for two place decimals.
5. Waveguide size equivalent to MIL-W-85/1-091.
6. DC polarity must be observed.

Ⓐ FIGURE 1. Switch configuration and schematic, part no. M55041/6-001 - Continued.

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Ⓐ TABLE I. Electrical performance characteristics.

Dash number	Figure number	Housing	Operating frequency range	Nominal operating voltage	Pull-in voltage (less than)	Dropout voltage (less than)	Operating current (max) 4/	Holding current (max)	Isolation (min)	VSWR (max)	Insertion loss	Switching time (max)
			GHz	V dc	V dc	V dc	amperes	amperes	dB		dB	ms
001	1	1/ 2/	11-17	28	20	10	1.5	.6	60	1.15:1 3/	0.2	100

TABLE I. Electrical performance characteristics - Continued.

Dash number	Waveguide pressurization	Life cycles (X1000)	Weight (max)
	lb_f/in^2		oz
001	20	100	21

1/ Enclosed, sand- and dust-proof when waveguide flanges are capped.

2/ Rotor and housing of aluminum alloy.

3/ VSWR 1.4:1 max from 11 to 12.4 GHz.

4/ At 28 V dc.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Power connectors: See figure 1.

Operating temperature range: 0°C to +60°C.

Electrical performance characteristics: See table I.

Drive voltage: 22 - 30 volts.

RF power handling capability: Average - 5 W; peak - 1 kW with a duty cycle of 5×10^{-3}.

Part number: M55041/6-001.

Ⓐ Substitutability data: The switch covered by this specification sheet is substitutable for the user's or manufacturer's part numbers specified in table II. This information in no way implies that the manufacturer's part number or NSN is suitable as a substitute for the military part number.

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TABLE II. Substitutability data.

Dash number	Manufacturer's code	Commercial type	NSN 5985
001	05869	712401-1	---
001	82152	33C12500	---

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, SM
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - AS, MC, OS, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0975-3)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-55041/6A	Switches, Waveguide, 1P2T, Fail-safe

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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